

AGRICULTURE PERFORMANCE ANALYSIS

Major Project Report

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Academic Project | Data Visualization & Analytics

Abstract

This project analyzes a farm-level agriculture dataset containing 4,000 records and 28 variables. The dataset covers location, crop, season, farm area, weather, soil properties, fertilizer and pesticide usage, seed quality, irrigation method, yield, production, market price, cost, revenue, profit, water use, water efficiency and disease/pest risk. The analysis uses descriptive statistics, grouped comparisons, correlation analysis and visualizations to identify patterns relevant to agricultural performance. The overall average yield is **5.27 tonnes/ha**, average profit is **■111,556**, and about **50.8%** of farms have positive profit in the supplied dataset.

Keywords: Agriculture Analytics, Data Visualization, Crop Yield, Profitability, Irrigation, Water Efficiency, Python

1. Introduction

Agriculture performance is influenced by environmental conditions, farm inputs, crop choice, irrigation practices and market economics. Raw farm records can contain enough information to compare performance across crops, seasons, states and irrigation methods. This project converts the supplied CSV dataset into interpretable summaries and visual evidence.

2. Problem Statement

Farm-level data is often difficult to interpret without systematic analysis. The project addresses the need to identify which crop, season, location and irrigation groups show stronger yield and profit, while also examining water efficiency and disease/pest risk. The aim is descriptive decision support rather than a claim of causal relationships.

3. Objectives

- Load and inspect the agriculture dataset.
- Summarize crop, season, state and irrigation performance.
- Compare yield, profit and water-efficiency indicators.
- Examine relationships between numerical variables using correlation.
- Create clear visualizations suitable for presentation.
- Identify practical areas for future predictive and dashboard-based work.

4. Dataset Description

The supplied dataset contains **4,000 farm records** and **28 columns**. It includes 8 states, 8 crop categories, 3 seasons and 4 irrigation methods.

Category	Values / range
Records	4,000
States	Andhra Pradesh, Gujarat, Karnataka, Madhya Pradesh, Maharashtra, Punjab, Tamil Nadu, Telangana
Crops	Chilli, Cotton, Groundnut, Maize, Pulses, Rice, Sugarcane, Wheat
Seasons	Kharif, Rabi, Zaid
Irrigation	Drip, Flood, Rainfed, Sprinkler
Yield	0.30–101.44 t/ha
Profit	■-1,019,223 to ■4,345,421

5. Methodology

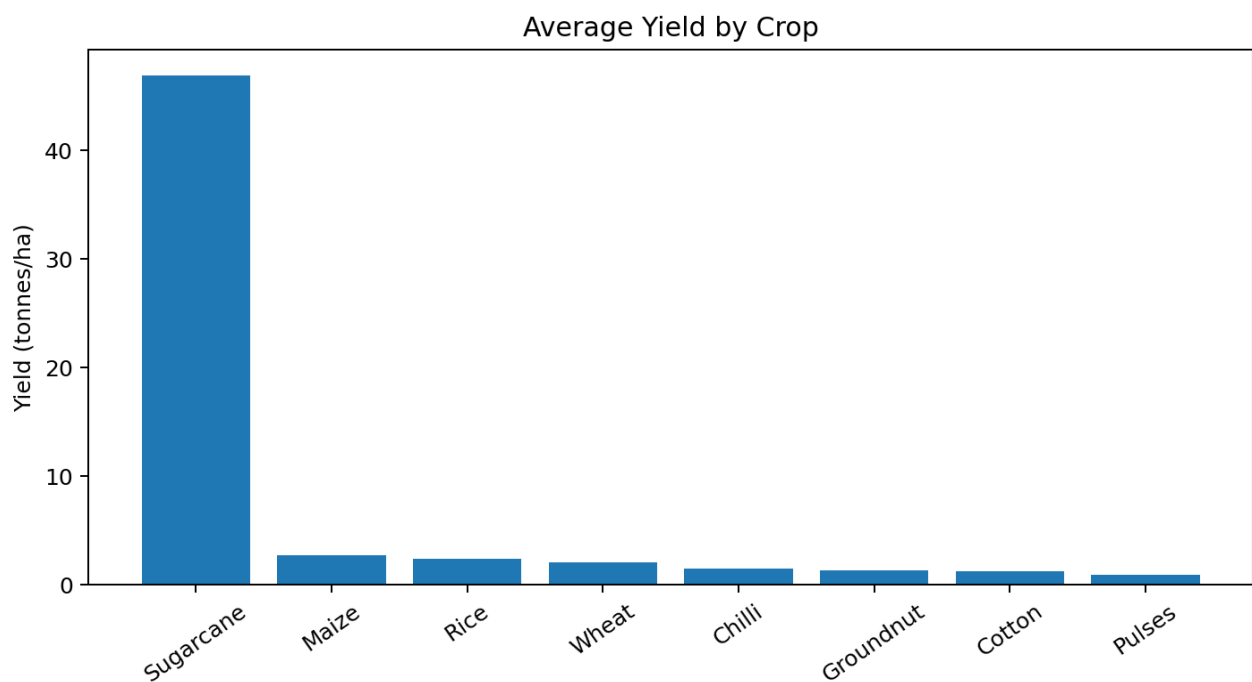
The analysis follows a simple exploratory data-analysis workflow:

- Data loading with Pandas from CSV.
- Data-type and completeness inspection.
- Descriptive statistics for key numerical fields.
- Group-by aggregation by crop, season, state and irrigation method.
- Correlation analysis among numerical indicators.
- Visualization using bar charts, histogram and scatter plot.
- Interpretation of patterns with attention to the descriptive nature of the dataset.

6. Results and Analysis

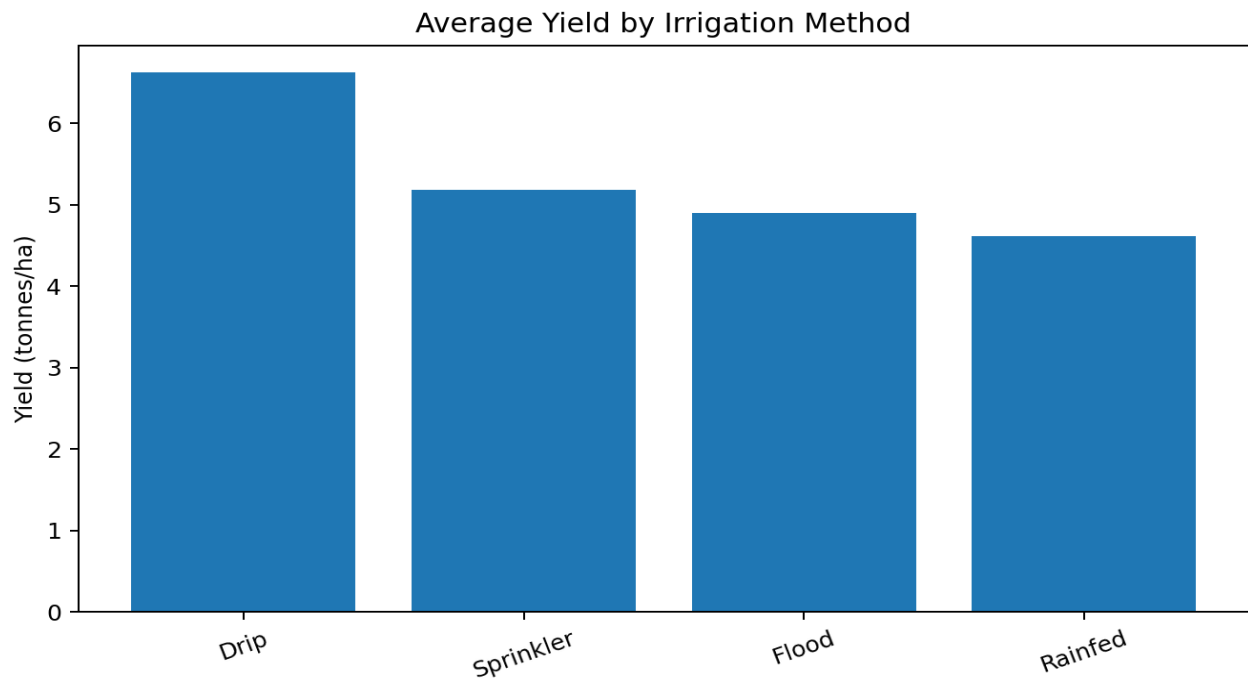
6.1 Crop Performance

The dataset shows substantial differences in average yield across crops. Sugarcane has the highest average yield at **46.94 t/ha**. Pulses has the lowest average yield at **0.92 t/ha**. Because yield alone does not represent economic performance, the project also compares profit and revenue.



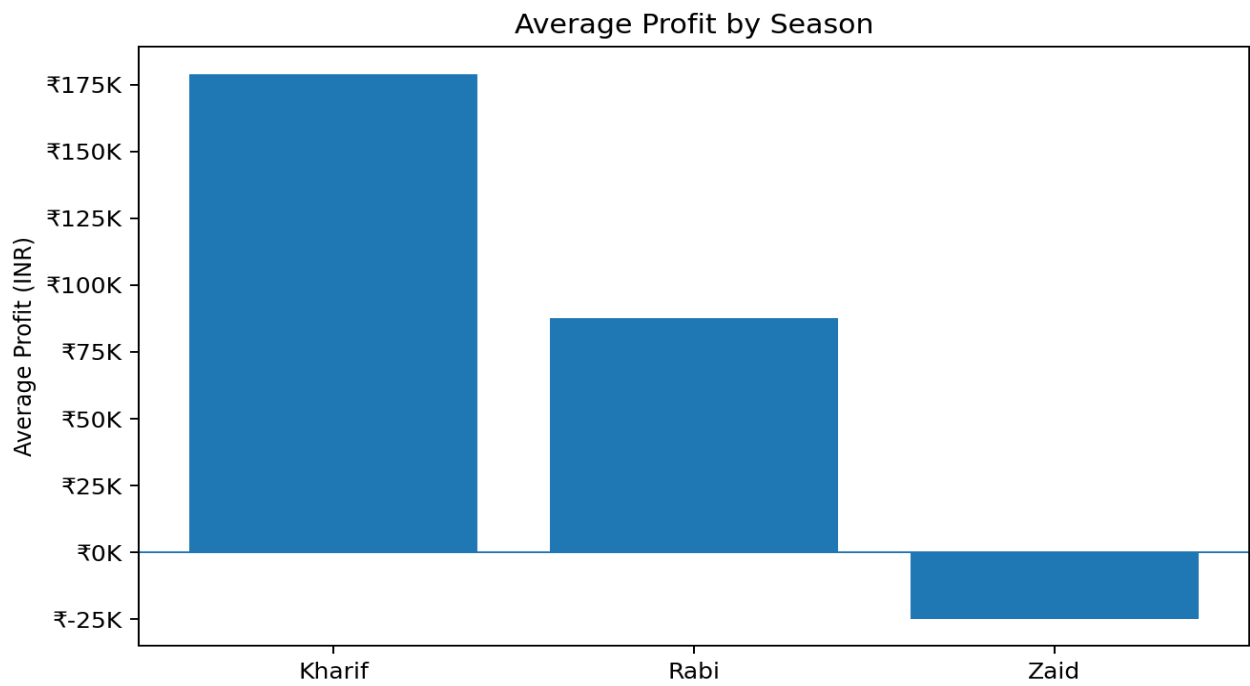
6.2 Irrigation Performance

Drip irrigation records the highest average yield at **6.62 t/ha** and the highest average profit at **■219,626**. Flood irrigation has lower average water efficiency than the other methods in this dataset. These results are descriptive associations and do not establish that irrigation method alone caused the observed outcomes.



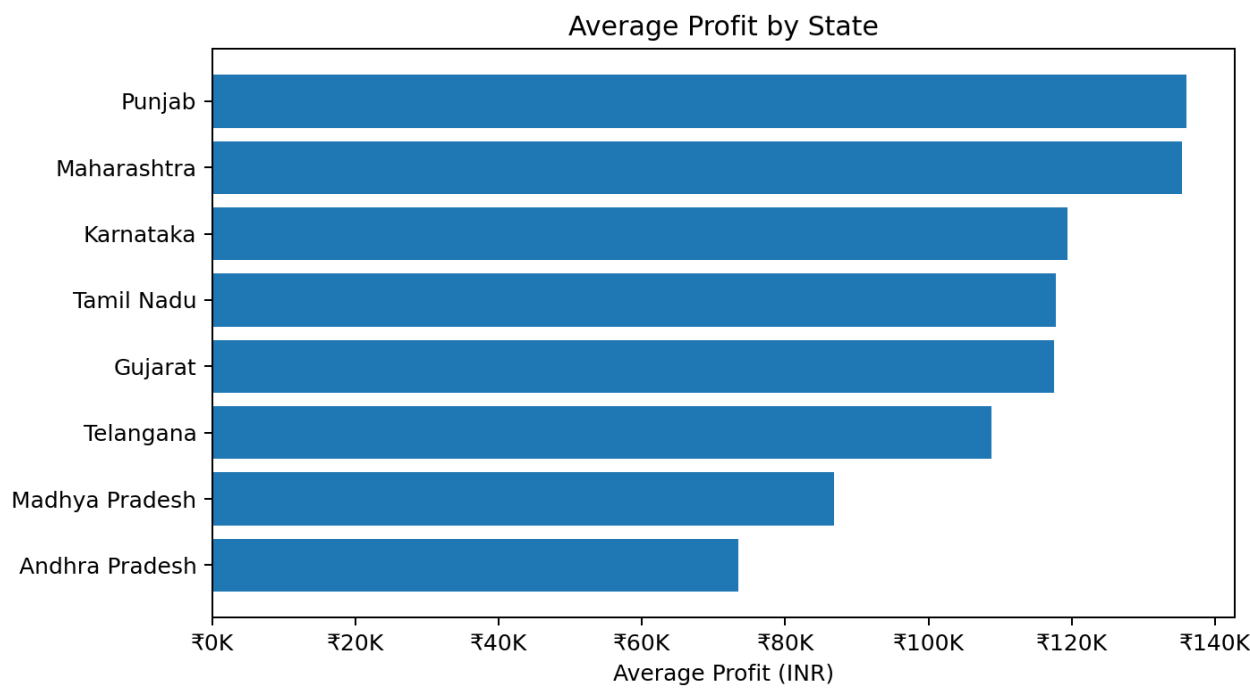
6.3 Seasonal Performance

Kharif has the highest average yield (**5.64 t/ha**) and the highest average profit (**₹178,915**). Zaid has an average profit of **₹-24,805**, which is negative in the supplied data.



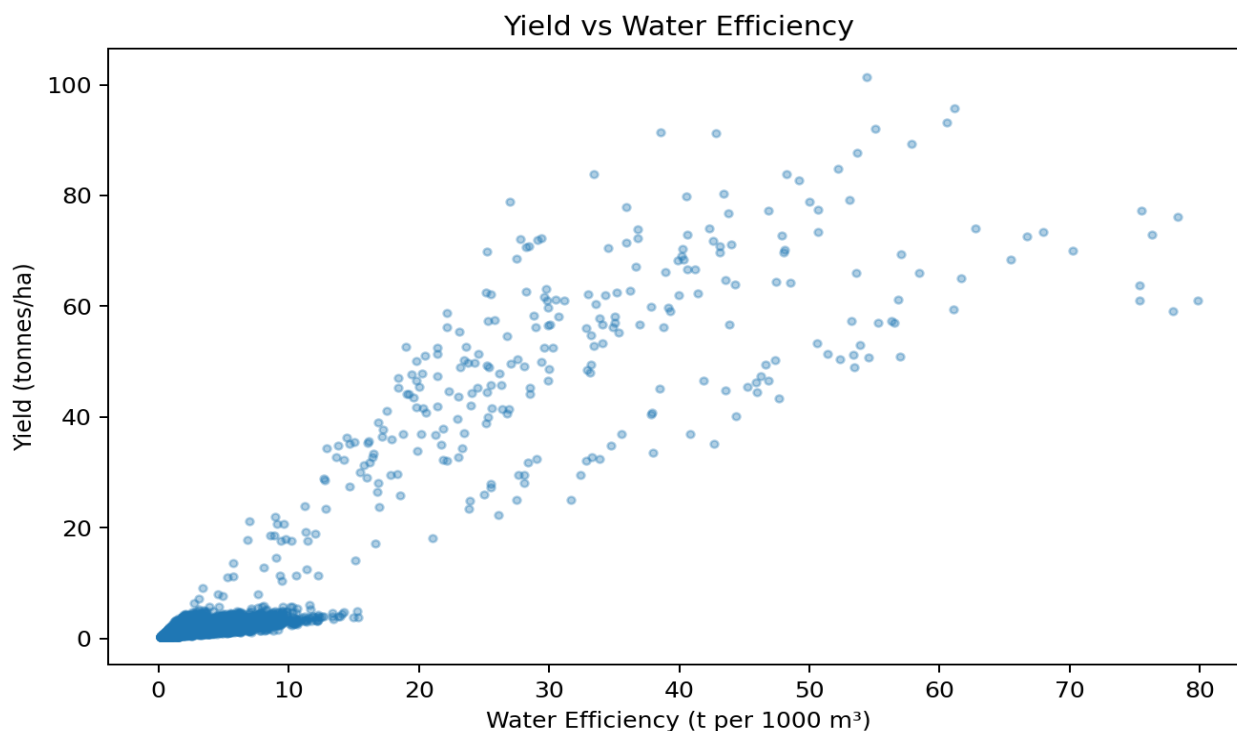
6.4 State-wise Performance

Punjab has the highest average profit among the states at **₹136,026**, while Andhra Pradesh has the lowest at **₹73,454**. Differences may reflect the combination of crop mix, farm size, input costs, prices and other variables in the records.



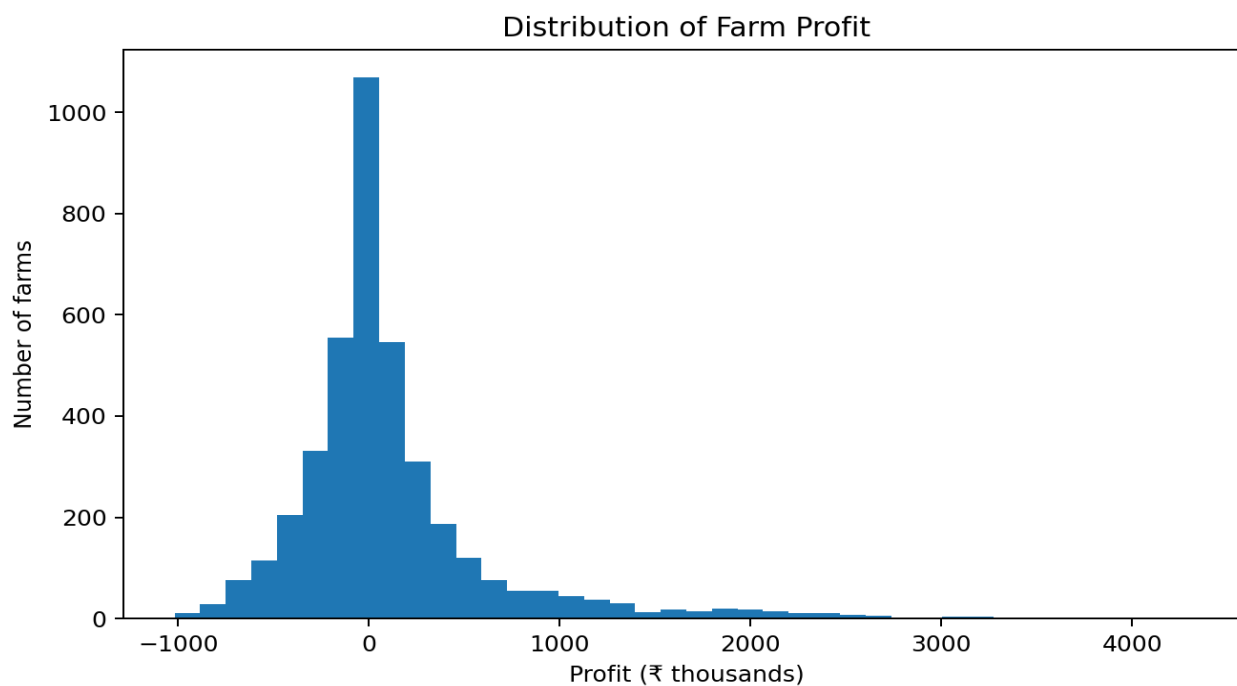
6.5 Water Efficiency

Water efficiency has a strong positive correlation with yield in the supplied dataset ($r = 0.92$). The overall average water efficiency is **5.39 tonnes per 1,000 m³**. The scatter plot provides a visual view of this relationship.



6.6 Profit Distribution

Average profit is **₹111,556**, but the distribution is wide. Approximately **50.8%** of records have positive profit, showing that average performance alone can hide substantial variation between farms.



7. Key Findings

- Dataset size: 4,000 farm records across 8 states.
- Average yield: 5.27 tonnes/ha.
- Average profit: ₹111,556.
- Highest average crop yield: Sugarcane (46.94 t/ha).
- Highest average irrigation-method yield: Drip (6.62 t/ha).
- Highest average seasonal profit: Kharif (₹178,915).
- Highest average state profit: Punjab (₹136,026).
- Yield and water efficiency correlation: 0.92.

8. Limitations

- The dataset is analyzed as supplied; no external validation is performed.
- The analysis is descriptive and does not prove causal effects.
- Some values show large differences across crop categories, so comparisons should consider crop type and scale.
- No predictive model or real-time data integration is included in the current version.

9. Future Scope

- Build machine-learning models for yield and profit prediction.
- Develop crop recommendation using soil and weather variables.
- Add disease/pest risk classification and early-warning alerts.
- Create an interactive Streamlit or Power BI dashboard.
- Integrate historical weather and market-price time series.
- Add optimization for irrigation and input allocation.

10. Conclusion

The Agriculture Performance Analysis project demonstrates how farm-level data can be transformed into useful visual insights. The supplied records show clear differences in crop, irrigation, seasonal and state-level performance, while water efficiency is strongly associated with yield. The resulting analysis can serve as a foundation for a future decision-support dashboard or predictive agriculture system.

11. Tools Used

Python, Pandas, NumPy, Matplotlib, Jupyter Notebook / VS Code, CSV dataset.

12. Project Files

Presentation: Agriculture Performance Analysis – Major Project PPT. GitHub repository: To be added by the student.